

ADR-003 — Exploitability is the headline metric, and VGC is a poker problem

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(Written at 3.57.1 and accepted the same night. 3.59.0 is the release that PUBLISHED it — the white paper, the deck, the technical documentation, docs/SUMMARY.md , docs/MODELS.md , ADR-001, ADR-002, docs/ROADMAP.md §5 and docs/MEASURE.md all now carry this decision. The content below is unchanged; a decision that lives only in its own ADR has not landed.)

Status

Accepted. Supersedes nothing; **promotes** docs/POKER-TO-POKEMON.md from one track among several to the project's central claim, because the evidence it was missing now exists.

Context

Will, 2026-08-06: "OUR EDGE CAN BE SEVERAL THINGS, OUR OWN MEDICHAM, THE POKER ASPECT OF BEING UNEXPLOITABLE MIX."

docs/POKER-TO-POKEMON.md already argued that VGC is formally the same object as heads-up poker — a two-player, zero-sum, imperfect-information game — and that the correct solution concept is therefore **a Nash equilibrium in mixed strategies, not a single best move**. It argued this from theory. It had no measurement of what happens to a project that assumes otherwise.

VGC-Bench supplies that measurement. Angliss, Cui, Hu, Rahman and Stone (AAMAS 2026) trained behaviour cloning on 700k human logs and fine-tuned with PPO under self-play, fictitious play and double oracle. In a single-team mirror match their agent beat a World Championships competitor. And:

- "In almost all cases, **all agents are approximately 100% exploitable**" — measured by training a best-response policy against each agent.
- Their expert tester's feedback: "although the agent is strong on initial play, it does have noticeable dips in performance in certain states. **After enough successive games, strong human players can adapt and beat the agent.**"
- Against their **advanced** (not expert) tester, the agent won **2 of 5**.

That is not a weakness of their execution. It is the predicted behaviour of a **compiled policy** — a fixed map from state to action — in an imperfect-information game. A best response can find and drill its blind spots, and so can a human given five games. Poker learned this over 2007–2021 and answered it with equilibrium mixing and continual re-solving (CFR, DeepStack, Libratus, ReBeL).

Two of their five stated open problems are **search-based planning** and **opponent modelling**. Nobody in the field has done either in VGC doubles.

AND WE ALREADY RAN THIS EXPERIMENT ON OURSELVES AND READ IT AS A WIN. Will, 2026-08-06, on being shown the GTO/exploitative distinction: "IS THAT WHAT THE GREEDY IS" — yes, exactly.

docs/SUMMARY.md:105 records "take the best move instead of sampling it — **+12 points, 79.7% of decisive pairs**", and it shipped. CHANGELOG.md:2739 even notes it in passing: "greedy-over-sampling shipped for +12 points and makes the policy **more exploitable**." We measured the poker trade-off, wrote down both halves, and took the exploitative branch — **because the metric was win rate**.

Against a self-play opponent pool, argmax wins by 12 points. Against a best-response exploiter, a deterministic policy is a lookup table to memorise. Same experiment, opposite verdict, and the only thing that changed is what we chose to measure. This is the cleanest internal evidence for the decision below.

One caveat so the reverse is not overclaimed: `greedy=false` is a mixed strategy but it is **the wrong mixture**. A softmax over learned scores randomises in proportion to *how good the scorer believes each move is*; an equilibrium mixture randomises so the opponent *cannot punish you*, and is determined by the payoff matrix rather than by our confidence. Sampling is directionally right and is still not GTO. Closing that gap is what promoting SLOWKING means in practice.

Decision

- 1. Exploitability, not win rate, is this project's headline metric.** WOBBUFFET moves from side-check to primary instrument. The published comparator is VGC-Bench's ~100%.
 - 2. The thesis is that a re-solving agent should be harder to exploit than a compiled one.** A learned policy *recalls*; a search *recomputes*. A best-response exploiter attacks a fixed mapping, and a per-turn re-solve presents none. **Whether this survives simultaneity, stochasticity and a 6-turn horizon is UNKNOWN — that is the experiment, not the assumption.**
 - 3. MEDICHAM's justification is now explicit and falsifiable.** VGC-Bench used real Showdown via poke-env and carried no engine-correctness debt, because behaviour cloning and PPO do not need a fast simulator. We wrote one so that **per-turn re-solving is affordable**. Therefore: *the engine work is justified if and only if search pays*. ROADMAP #62 stops being a MILTANK question and becomes the project's gate.
- Supporting evidence that this is the real trade and not a rationalisation — every project that searches hits the engine-speed wall:

project	searches?	engine	depth reached
VGC-Bench	no	real Showdown	n/a
Future Sight AI	yes	modified Showdown	~3 turns in 15 s
Foul Play	yes	built poke-engine	~10+ turns

- 4. If search does not pay, adopt their method — and say so.** BC + PPO with self-play/FP/DO is published, open-source and reproducible. The fallback is cheap precisely because they made it so, and taking it would be a result, not a defeat.

Consequences

What this makes true

- The comparison with VGC-Bench does **not** require playing their agent, which is fortunate, since their checkpoints are Reg M-A and we are Reg M-B, and their own paper shows policies do not transfer across team sets. **Exploitability is intrinsic** — measured against a best-response trained against *you*, in *your* format — so the numbers are comparable although the agents can never meet.
- SLOWKING stops being "the preview solver" and becomes the shape of the whole agent.
- HYPNO (#52, deferred) gains a reason to stay deferred: Future Sight AI programmed play-by-opponent-rank and got a ~50% win rate against *every* rank — it learned to match its opponent instead of beating them. Modelling the opponent is fine; modulating our own strength is the trap.

What this costs

- Exploitability is expensive to measure. Each number needs a best-response trained against a frozen agent, which is why `data/exploitability.json` is currently VOID and ABRA has **no** exploitability number at all.

Producing a trustworthy one is now a first-class deliverable rather than a nice-to-have.

- It raises the bar on the frozen-release discipline: an exploitability run that moves under itself is the 2026-08-04 void repeating, and that void was an exploitability run.

What we are NOT claiming

- Not that we will beat VGC-Bench. Their agent beat a Worlds competitor; ours has never played a human.
- Not that search is known to work here. Metamon (RLC 2025) reached top 10% in singles with **no search at all**, and Future Sight AI removed its machine learning after finding a structural method beat it on both accuracy and speed. Both are live counter-evidence and both are recorded in `docs/PRIOR-ART.md`.
- Not novelty on the format or the infrastructure. VGC-Bench owns both, including the poke-env doubles support the field now uses.

The honest contribution, if the thesis fails

The instrumentation stands either way. No project in `docs/PRIOR-ART.md` publishes a mechanics census that must be shown RED before it counts, a step-level protocol differential against the official engine, or ratchets on silent failure — and none of them publishes what it retracted. If search loses, this is still the only account of what a hand-written VGC simulator gets wrong and how you would know, which is publishable precisely because everyone else avoided the problem by not having it.